

Deep Learning Practical Simple Explanations + Easy Google Colab Codes

1. Loading Dataset into PyTorch/Keras & Creating Train-Test Split

Simple Explanation

This practical teaches how to load datasets and divide them into training and testing data for model learning and evaluation.

```
from sklearn.model_selection import train_test_split  
from tensorflow.keras.datasets import mnist
```

```
# Load dataset
```

```
(X_train, y_train), (X_test, y_test) = mnist.load_data()
```

```
print("Training Shape:", X_train.shape)
```

```
print("Testing Shape:", X_test.shape)
```

2. Creating Functions to Compute Various Losses

Simple Explanation

Loss functions calculate prediction errors between actual and predicted outputs.

```
import tensorflow as tf
```

```
y_true = [1, 0, 1]
```

```
y_pred = [0.9, 0.2, 0.8]
```

```
# MSE Loss
```

```
mse = tf.keras.losses.MeanSquaredError()
```

```
print("MSE:", mse(y_true, y_pred).numpy())
```

3. Feeding Data to Pretrained Neural Network

Simple Explanation

Pretrained models like MobileNet are used for prediction without training from scratch.

```
from tensorflow.keras.applications import MobileNetV2  
  
# Load pretrained model  
  
model = MobileNetV2(weights='imagenet')  
  
print("Model Loaded Successfully")
```

4. Classifying Dataset Using CNN

Simple Explanation

CNN is used for image classification using convolution and pooling layers.

```
import tensorflow as tf  
  
from tensorflow.keras import layers, models  
  
model = models.Sequential([  
    layers.Conv2D(32, (3,3), activation='relu', input_shape=(28,28,1)),  
    layers.MaxPooling2D((2,2)),  
    layers.Flatten(),  
    layers.Dense(10, activation='softmax')  
)  
  
print(model.summary())
```

5. Classifying Data Using Pretrained Models / Transfer Learning

Simple Explanation

Transfer learning uses pretrained models to classify new datasets quickly.

```
from tensorflow.keras.applications import VGG16  
  
model = VGG16(weights='imagenet', include_top=False)  
  
print("Transfer Learning Model Loaded")
```

6. CNN for Automatic Digit Recognition

Simple Explanation

CNN recognizes handwritten digits using the MNIST dataset.

```
from tensorflow.keras.datasets import mnist
import matplotlib.pyplot as plt

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Display image
plt.imshow(X_train[0], cmap='gray')
plt.title("Digit: " + str(y_train[0]))
plt.show()
```

7. Object Classifier Using CNN in Python

Simple Explanation

CNN identifies and classifies objects from images.

```
import tensorflow as tf

model = tf.keras.Sequential([
    tf.keras.layers.Conv2D(32,(3,3),activation='relu',input_shape=(32,32,3)),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(10,activation='softmax')
])

print("Object Classifier Ready")
```

8. Training Popular Neural Networks on Custom Dataset

Simple Explanation

Popular models like ResNet and VGG are trained on custom datasets.

```
from tensorflow.keras.applications import ResNet50  
  
model = ResNet50(weights='imagenet')  
  
print("ResNet50 Loaded")
```

9. Reuters Dataset Using Deep Neural Network

Simple Explanation

Deep neural networks classify Reuters news articles into multiple categories.

```
from tensorflow.keras.datasets import reuters  
  
# Load Reuters dataset  
(X_train, y_train), (X_test, y_test) = reuters.load_data(num_words=10000)  
  
# Print dataset information  
print("Training Samples:", len(X_train))  
print("Testing Samples:", len(X_test))  
  
# Print first news label  
print("First Label:", y_train[0])
```